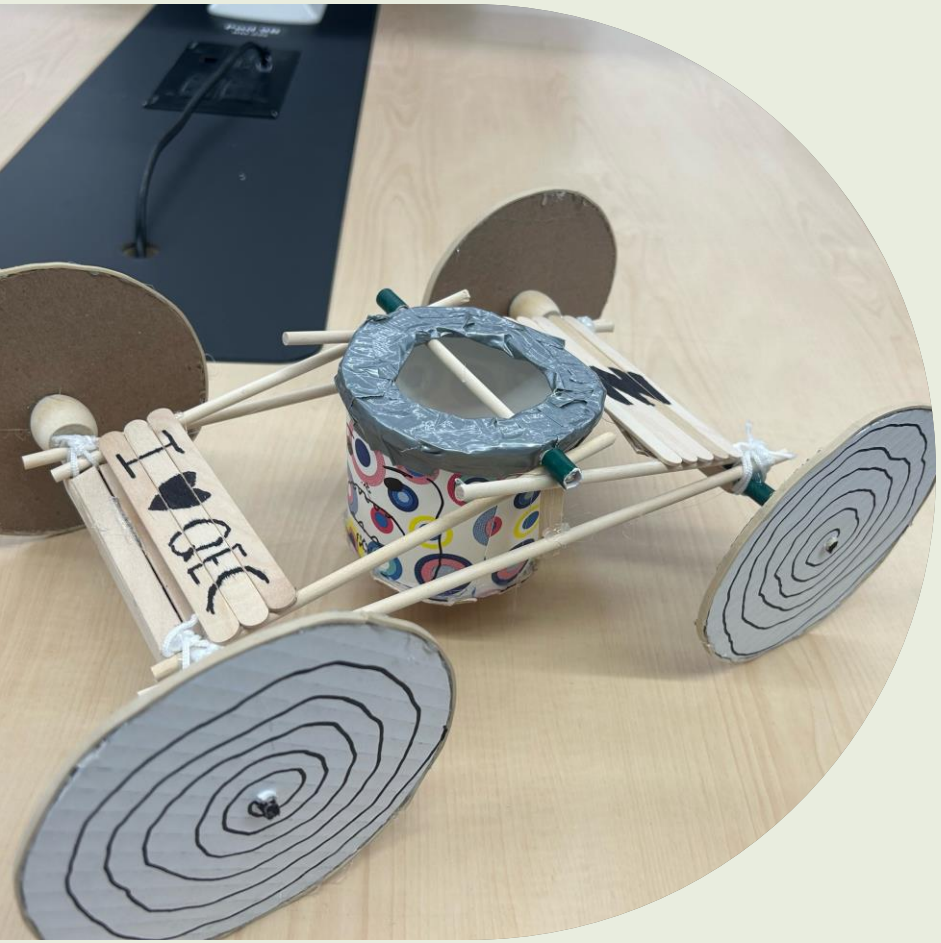


Water Transport
Small Vehicle

Thatch; A simple, accessible technology

Problem Statement

Water Transport for Rough and Inclined Terrain

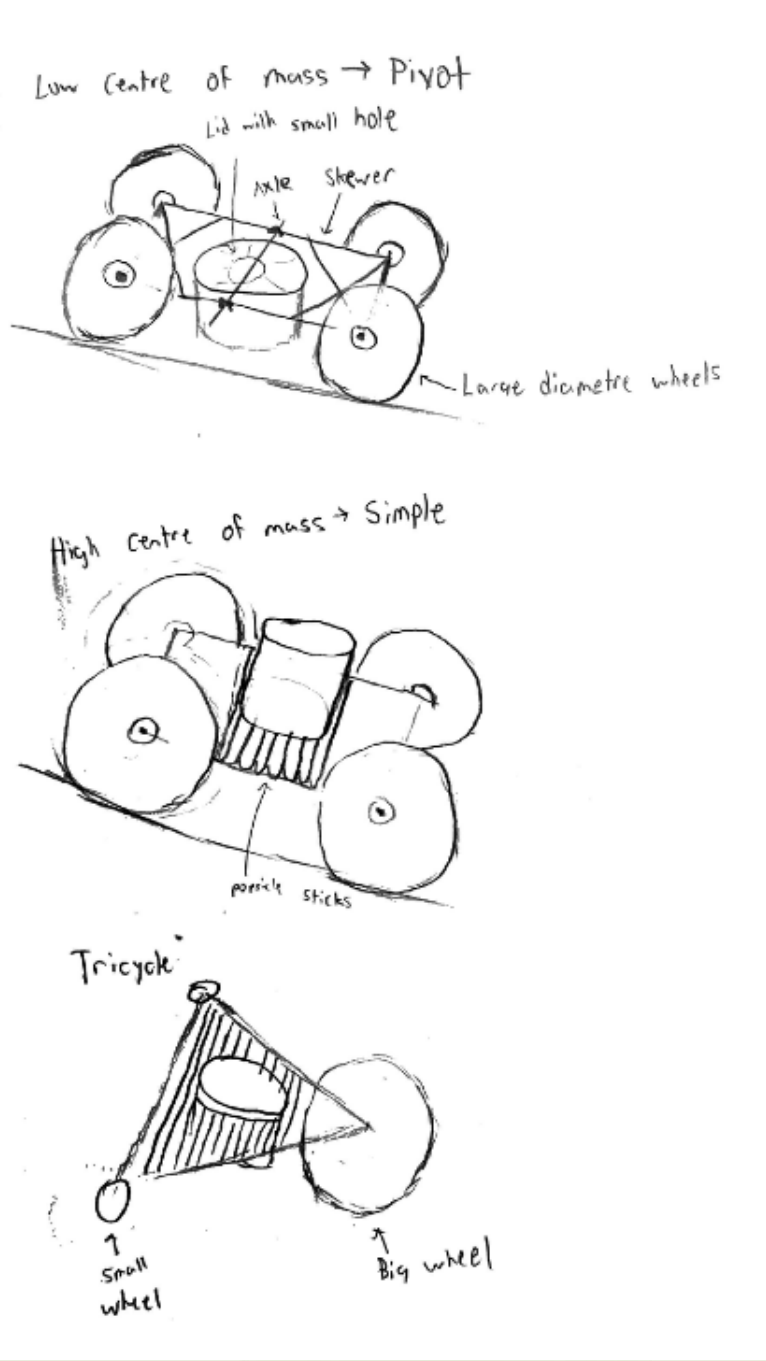
- Remote communities need safer, more efficient water-transport solutions
- Challenges: Rough and Inclined Terrain

- ClearPath: Non-Profit
- Design is scalable for real-world use
- Stakeholders: Remote communities

- Use only provided sustainable materials
- Top-fill water container
- Must carry ~75% container volume

GOAL:

- **Transport water over obstacles and rough terrain**
- **Minimize spillage**
- **Maximize travel distance**
- **Prototype: Simple and Accessible**



Our Goals









01

Simple

- Scalable
- Manufacturable

02

Accessible

- Financially Feasible
- Sustainable

Prototype: Stable and Manufacturable

Vehicle Trade Study

Figures of Merit	Factor	4 Wheel	Tricycle	Small Wheel	Large Wheel	High Mount	Mid Mount	Low Mount	Short Axle Separation	Long Axle Separation
Manufacturability	0.3	1	1	1	0.8	0.5	0.6	0.5	0.7	1
Stability	0.3	1	0.6	0.3	1	0.3	0.5	0.8	0.2	1
Water Spillage (M1, M2)	0.2	0.8	0.8	0.3	0.8	0.5	0.6	0.7	0.5	0.8
Rough Terrain (M2)	0.1	1	0.5	0	1	0.2	0.5	0.8	0.8	0.8
Bumpy Terrain (M2)	0.1	1	0.5	0	1	0.2	0.5	0.8	0.8	0.8
	TOTAL	**0.96**	**0.74**	**0.39**	**0.9**	**0.38**	**0.55**	**0.69**	**0.53**	**0.92**

Trade Studies

- 4 Wheel
- Large Wheel Diameter
- Low Mount
- Long Axle Separation

4 Wheel
Control

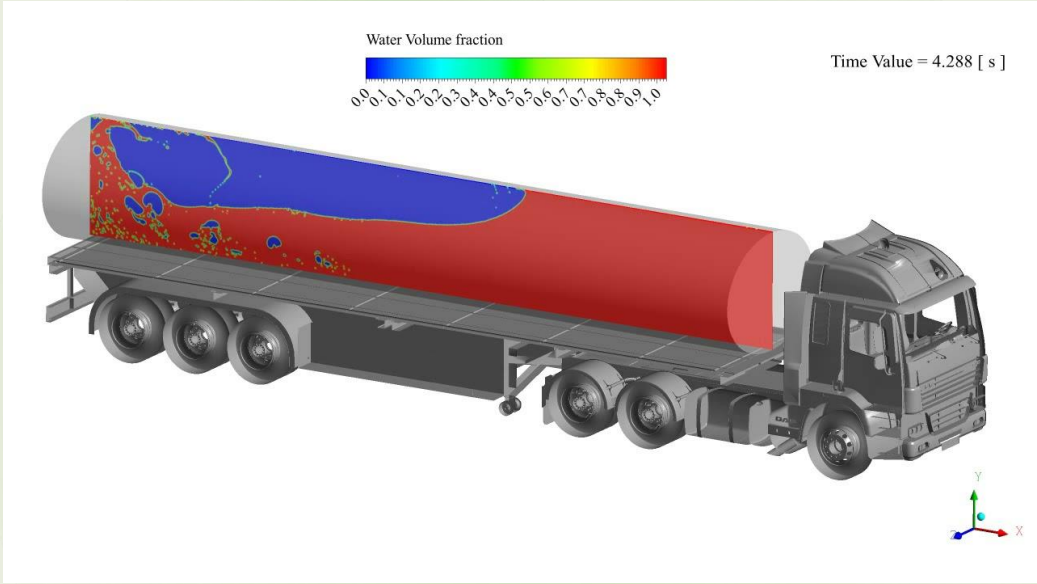
Low Mount
Low CoG

Large Wheel Diameter
Terrain

Long Axle Separation
Stability

Sloshing Effects

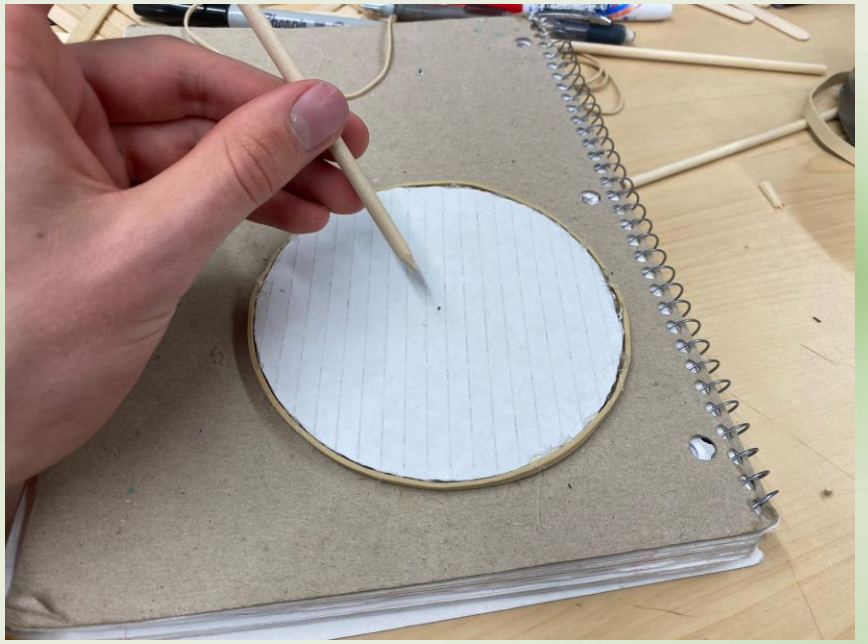
Water sloshing can shift the moment of inertia of the vehicle causing instability.



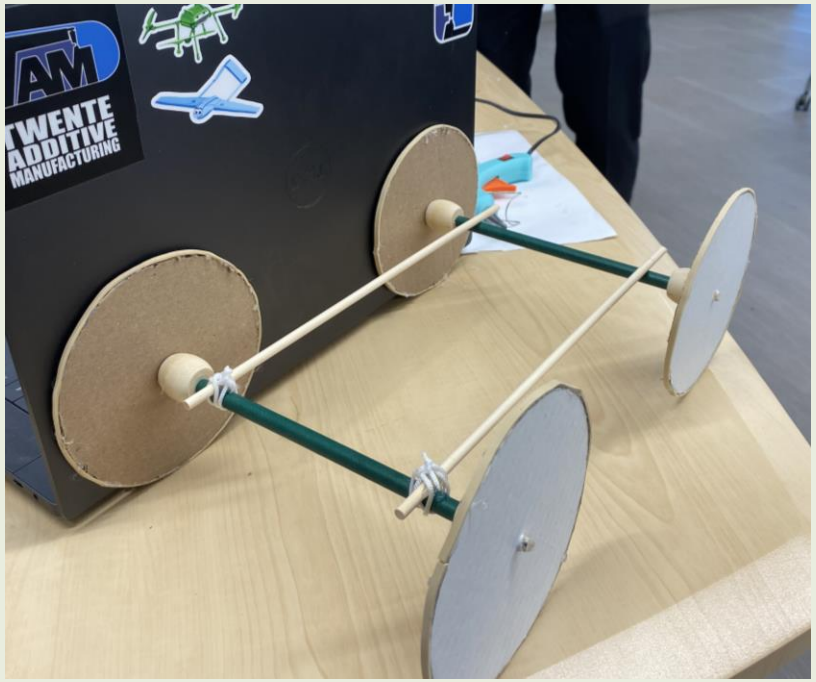
Our Solution:

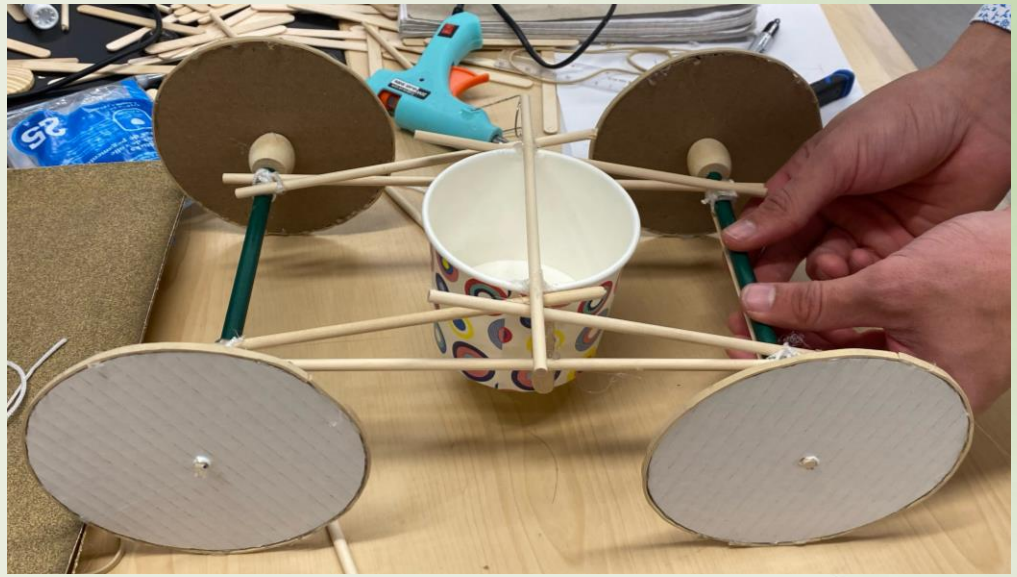
- Rubber bands to provide damping
- Constant level tilt mechanism

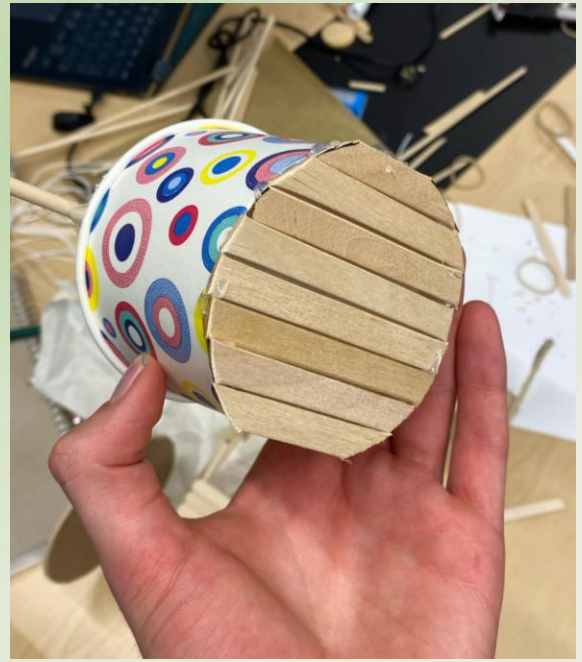

QEC






TWENTE
ADDITIVE
MANUFACTURING





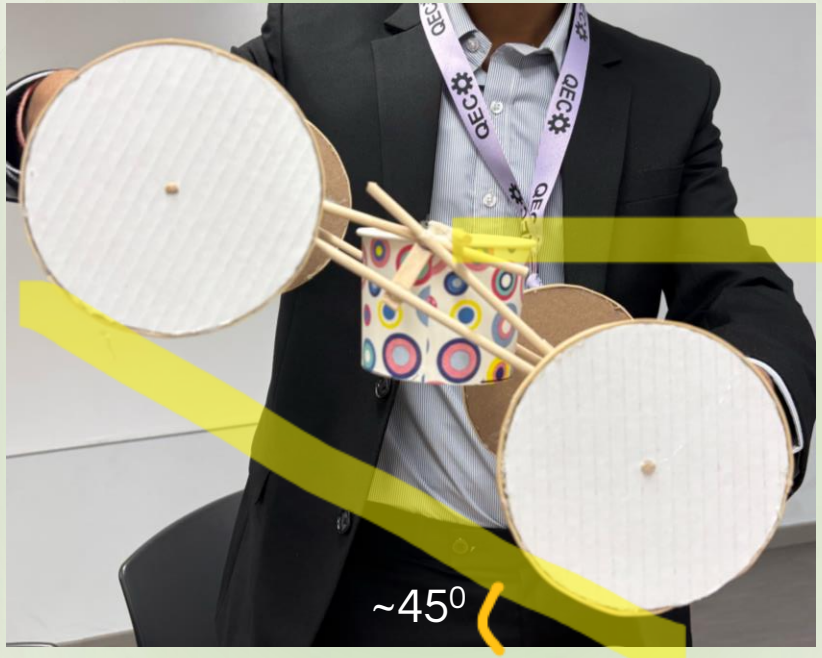
Iteration Preliminary Design Testing

Vehicle Alignment Testing

- Wheels and axles tested on smooth, level ground
- Straight string used as reference for expected travel path
- Vehicle consistently veered right → indicated axle misalignment
- Issue corrected by re-gluing and realigning axle

Tilt Testing

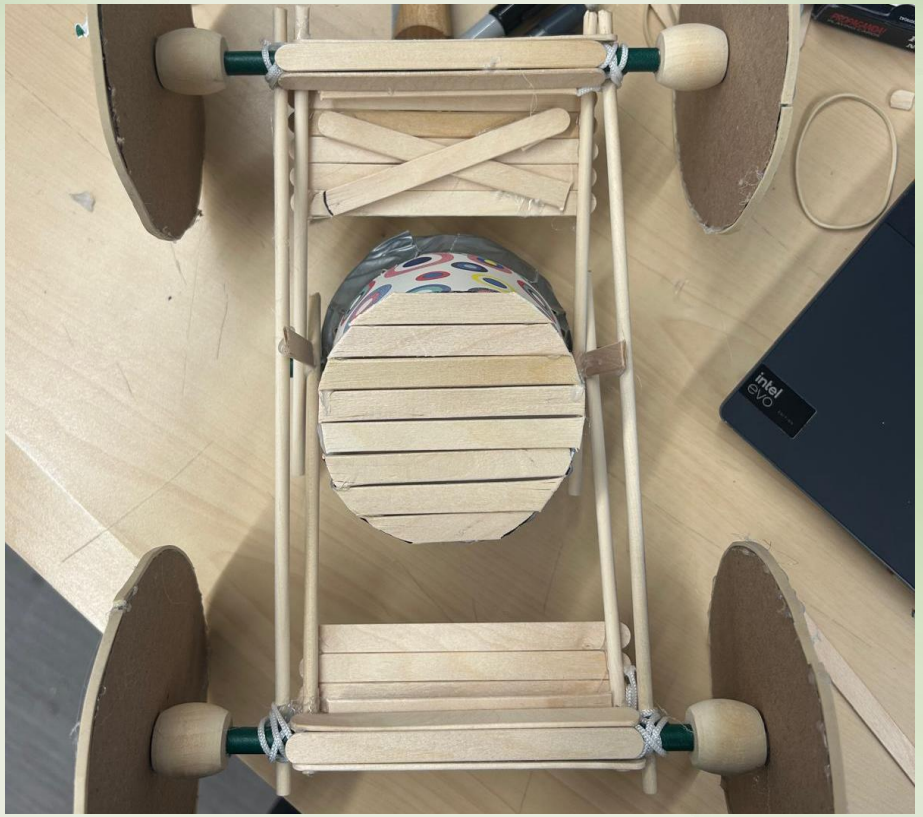
- Tilt test (≈45° incline) performed with empty cup
- Cup remained level → confirms effective stabilization design
- Abandoned Rubber Band Damping

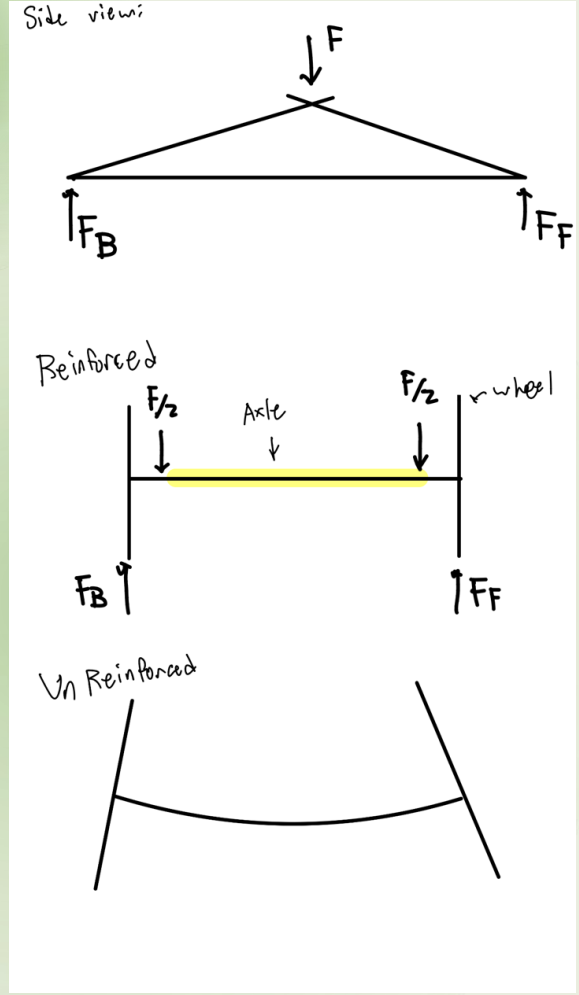


Iteration - Forces and Loading Test

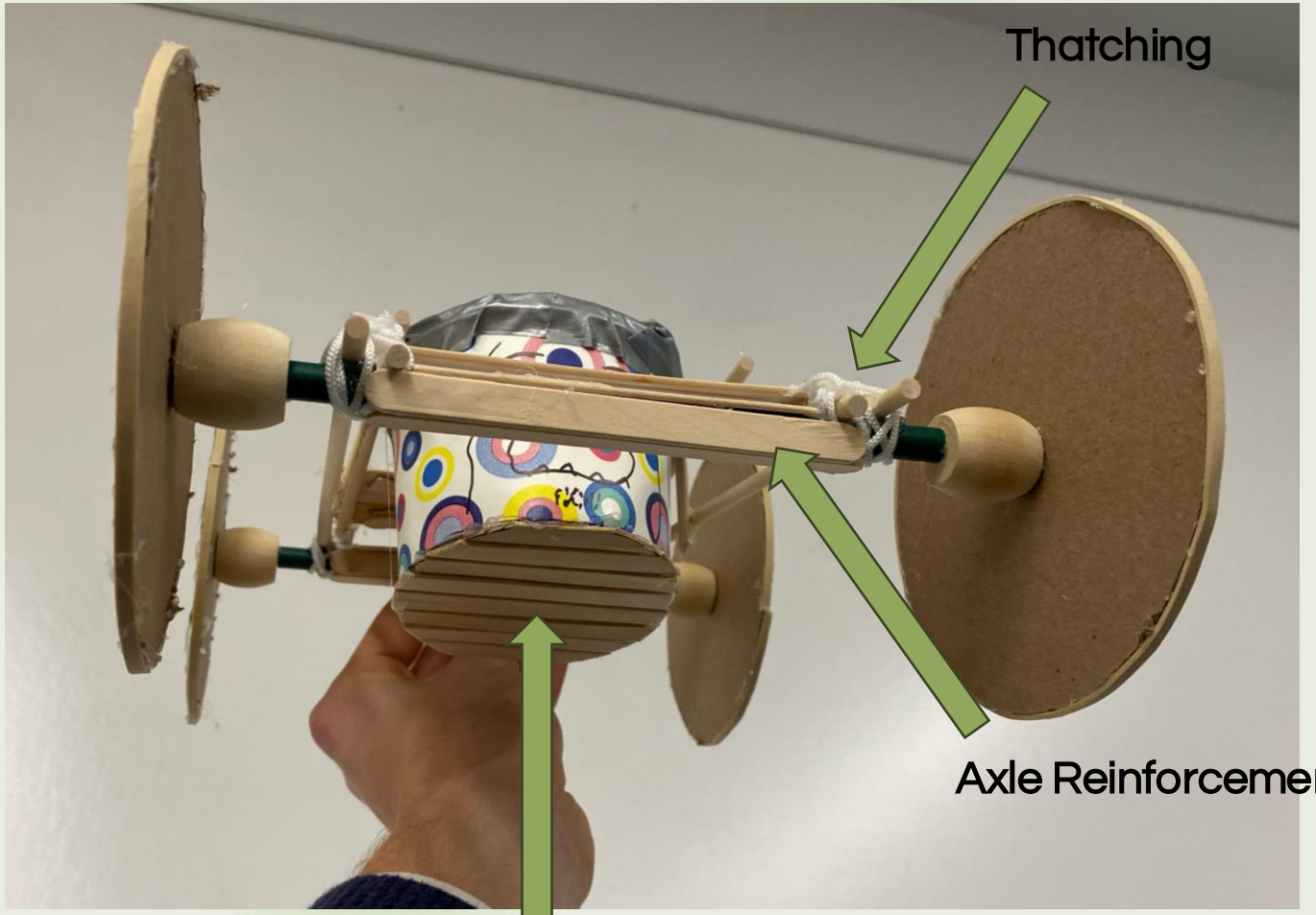
-Locations of forces within the structure were approximated to justify design choices

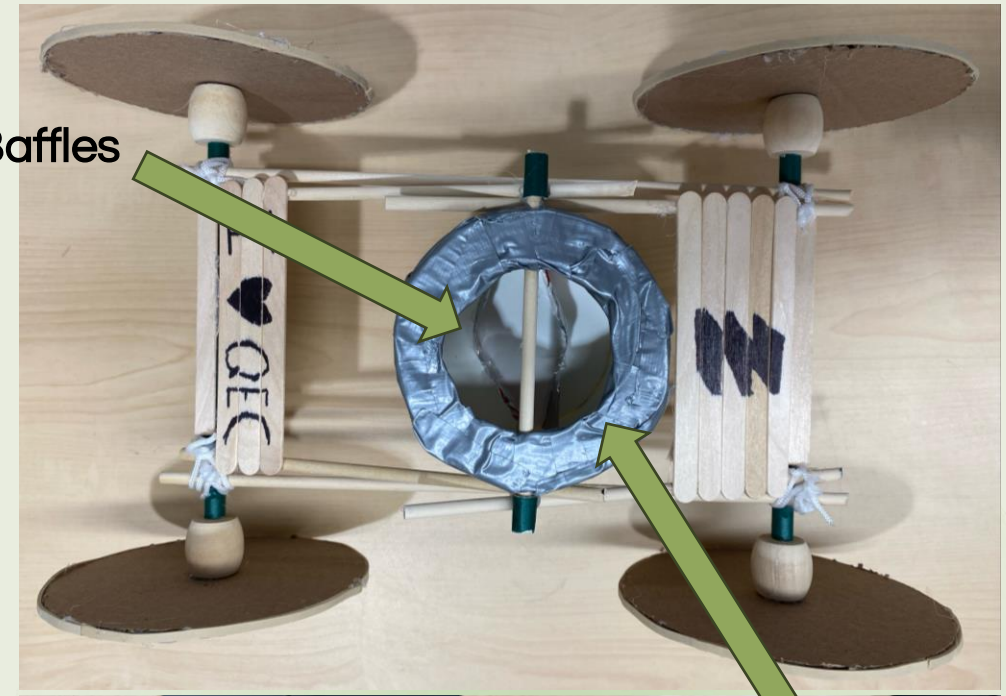
Loading Test:
-Vehicle loaded to over its max payload (500g)
-Structure inspected for damage and displacement

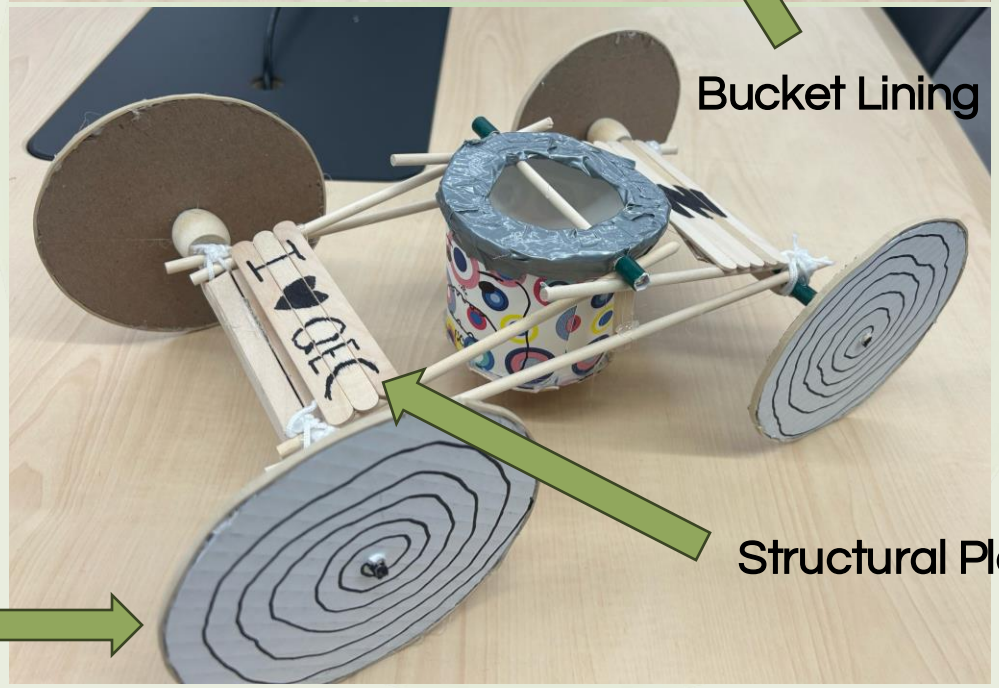




Final Design

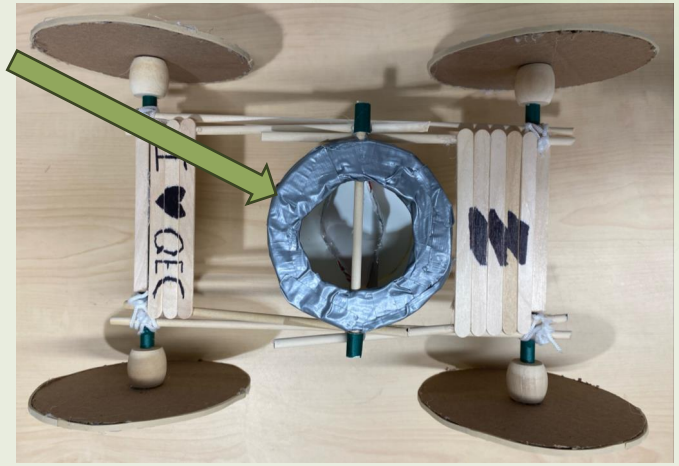






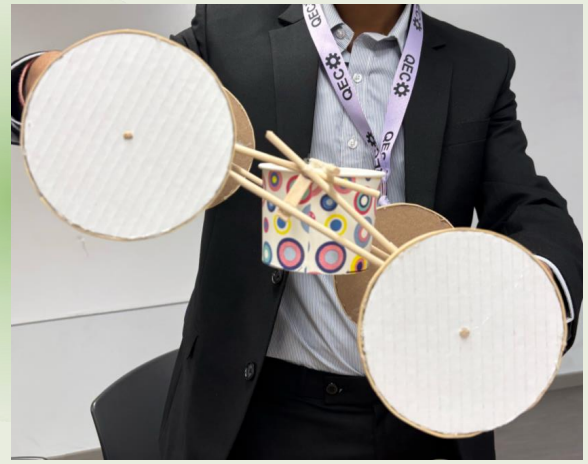
Final Design – Water Slosh Minimization

Baffles



Prototype Materials	Prototype Component
Cardboard	Wheel
Rubber Bands	Wheel Rim
Wood Dowels	Fuselage
Wood Stoppers	Axle
Paper Straws	Axle
Hot Glue	Assembly
Paper Cup	Water Container
Soda Can Metal	Water Container Baffles

Level Tilt Mechanism





Baffles are internal partitions that disrupt fluid flow in tanks

Prototype Improvement

And Implementation Strategies

01

Wheel Environment Limitations

- Wheels are too thin
- Performance Effects: limited functionality in rugged environments, i.e sand

Implementation

- Thicker wheels,
- TPU printed

02

Vertical Shock Absorption

- No vertical shock considerations
- Performance Effects: Would likely spill water in vertical drops.

Implementation

- Rubber Band Shocks
- Mounted to the Water bucket and frame

Scalability and Implementation

Water Output Mechanism

- Has no method of water output
  - Not accessible
  - Output tubing

Water Input Mechanism

- Has a limitedly closed surface
  - Cap or Screw on Mechanism

Financial Feasibility

- ✓ Simple, Minimal Parts
- ✓ Manufacturable, Pre-existing similarities
- ✓ Baffles like pre-existing fuel truck transportation baffles



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Drinking Water Advisories in over 100 Indigenous Communities in 2018*

Real World Applications

Remote Water Transportation

Remote Communities have unreliable access to potable water. This can be expanded to other liquids including fuel.

Disproportionally effects Indigenous Communities



Deliver potable water to remote communities

*https://www.theindigenousfoundation.org/articles/indigenous-safe-drinking-water-crisis-in-canada-overview